EXPERIMENT 4

String and Block Data Operations Using 8086 Microprocessor
Program A - To transfer a block of bytes using MOVSB with the REP prefix, including an overlapping block move controlled by the direction flag.
Program B – String Comparison with REPE CMPSB (start address 1100H)
Program C – Character Search with REPNE SCASB (start address 1200H)
Program D – Lower-case to Upper-case Using LODSB / STOSB (start address 1300H)

Objective:

To develop and verify **string and block data operations using the 8086 microprocessor**, including block data transfer, string comparison, character searching, and conversion of lower-case characters to upper-case characters using string manipulation instructions.

Algorithm:

- Initialize the required data and memory locations.
- **Program A:** Use MOVSB with the REP prefix to transfer a block of bytes. Use the direction flag to control the transfer direction during overlapping block movement.
- **Program B:** Store the strings starting from **1100H** and compare them byte by byte using REPE CMPSB until a mismatch or the end of the string is reached.
- **Program C:** Store the required string starting from **1200H** and search for the specified character using REPNE SCASB.
- **Program D:** Store the lower-case string starting from **1300H**. Use LODSB to read each character and convert it to upper-case, then use STOSB to store the converted character.
- Repeat the operations until the required number of bytes/characters are processed.
- Verify the output obtained from each program with the expected result.

A) Block Transfer using MOVSB with REP Prefix:

Code:

```
MOV SI,1000H
MOV DI,2000H
MOV CX,0004H
CLD
REP MOVSB
HLT
```

Input:

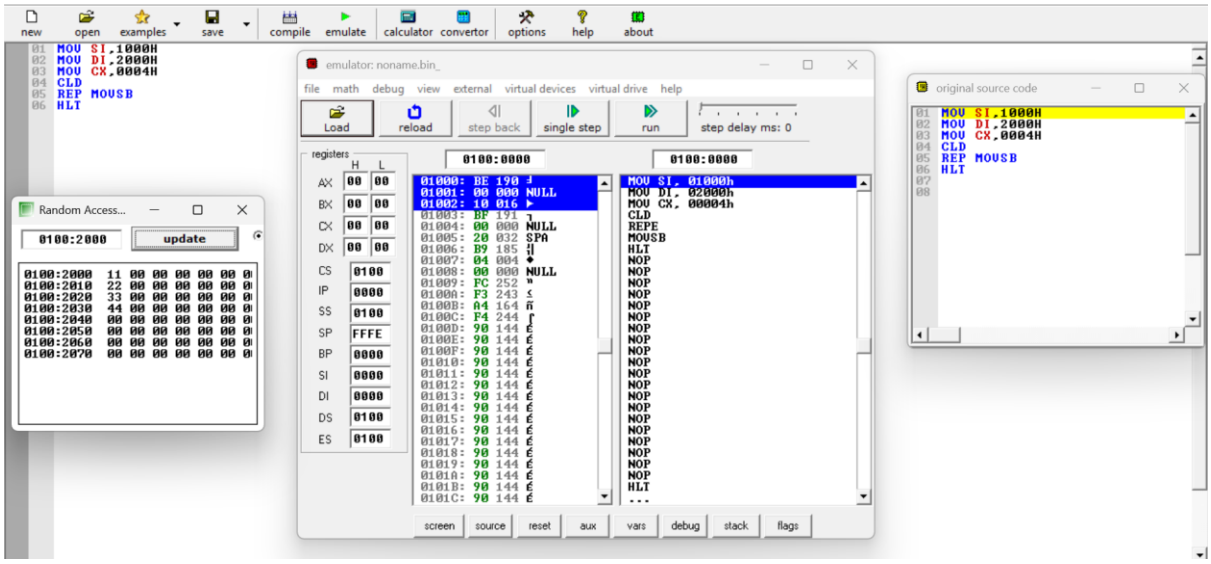
Input Address	Data

1000h	11h
1001h	22h

Output:

Output Address	Data
2000h	11h
2001h	22h

Output Screenshort:



B) String comparison using REPE CMPSB:

Code:

```
Mov SI,1400h;
Mov DI,1500h;
Mov CX,0005h;
CLD
REPE CMPSB;
HLT;
```

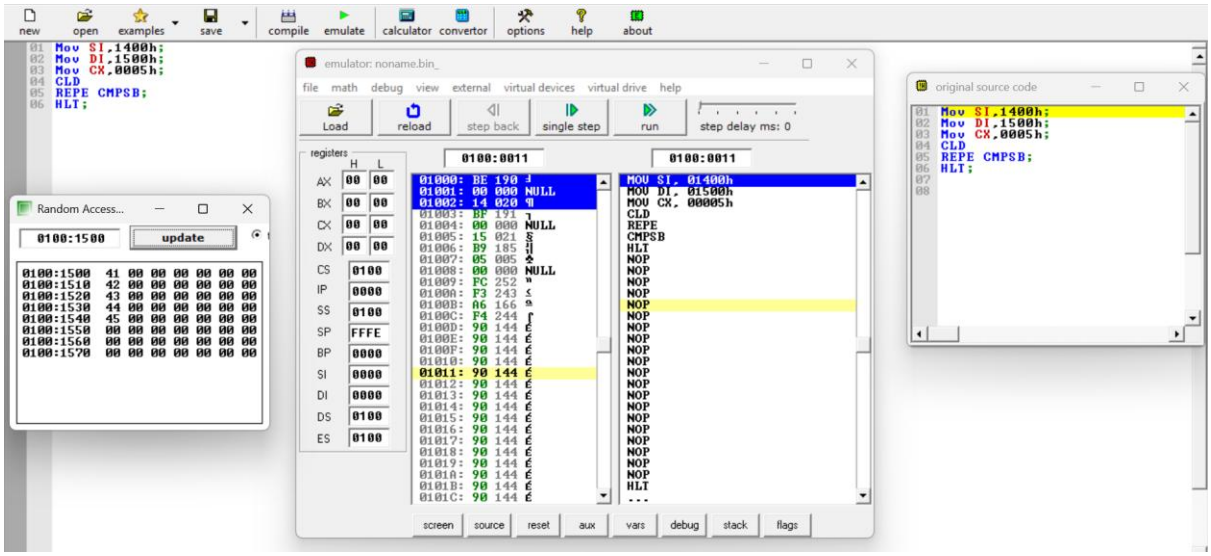
Input:

Input Address	Data
1400h	ABCD
1500h	ABCD

Output:

Output Address	Data
-	Equal
-	Equal

Screenshort:



C)Character Search using REDNE SCASB:

Code:

```
MOV DI,1600h;
MOV CX,0008h;
MOV AL,50h;
CLD
REPNE SCASB;
HLT;
```

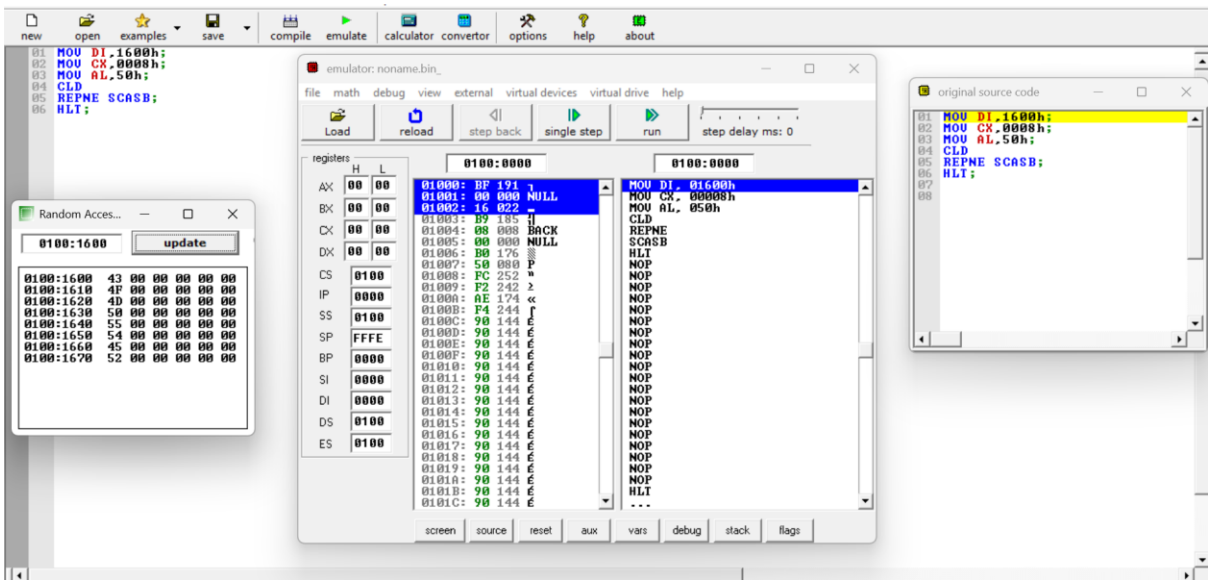
Input:

Input Address	Data
1600h	COMPUTER
-	Search=P

Output:

Output Address	Result
-	Found
-	Found

Output Screenshort:



D) Lower-case to Upper-case using LOBSBI STOSB:

Code:

```
MOV SI,1700H
MOV DI,1800H
MOV CX,0008H
CLD
BOAH:
LODSB
AND AL,0DFH
STOSB
LOOP BOAH
HLT
```

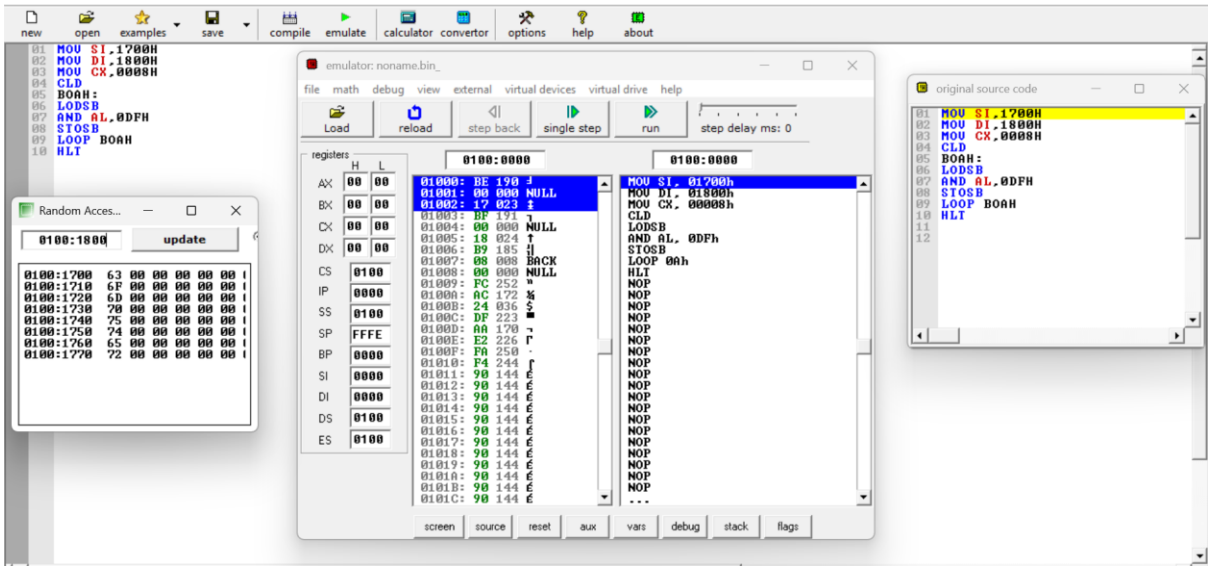
Input:

Input Address	Data
1700h	computer

Output:

Output Address	Data
1800h	COMPUTER

Student Screenshort:



Result:

Thus, the **block transfer, string comparison, character search, and lower-case to upper-case conversion** operations were successfully performed and verified using **8086 string and block manipulation instructions** such as MOVSB, REPE CMPSB, REPNE SCASB, LODSB, and STOSB.

Student Reflection:

I learned how to perform **string and block data operations using the 8086 microprocessor**. I understood the use of **string instructions, REP prefixes, direction flag, source and destination index registers**, and looping mechanisms. This experiment improved my understanding of **memory-to-memory data transfer, string comparison, character searching, and character conversion** in 8086 assembly language.